Experiment 2

Aim: Write a program to check whether a string belong to the grammar or not

Theory:

A Context-Free Grammar (CFG) is a formal way of defining the syntax of a language using a set of production rules that describe how strings can be generated from a start symbol. Each rule has the form A → α, where A is a non-terminal and α is a string of terminals and/or non-terminals, and rules can be applied repeatedly and in any order to derive valid strings of the language. Checking whether a given string belongs to a CFG means determining if there exists some sequence of rule applications, starting from S, that produces exactly that string. For simple grammars like the ones here, this membership can often be checked directly by recognizing the structural pattern the recursion produces, rather than building a full parse tree: for example, aSb-type recursion nested around a base case naturally generates strings with equal numbers of a's and b's, while symmetric recursion like aSa/bSb generates palindromes. CFGs form the basis of the syntax analysis (parsing) phase of a compiler, where the parser checks whether the token stream produced by the lexical analyzer conforms to the grammar of the programming language.

Practice Questions

1. Write a program to check whether a string belong to a grammar or not.

a) S -> aS

S -> Sb

S -> ab

String of the form : aab

b) S -> aSa

S -> bSb

S -> a

S -> b

The Language generated is : All Odd Length Palindromes

c) S -> aSbb

S -> abb

The Language generated is : anb2n , where n>1

d) S -> aSb

S -> ab

The Language generated is : anbn, where n>0

Code:

```
#include <iostream>
#include <string>
using namespace std;

bool checkA(string s) {
   int n = s.length();
   int i = 0;
   while (i < n && s[i] == 'a') i++;
   if (i == 0) return false;
   int firstB = i;
   while (i < n && s[i] == 'b') i++;
   if (i != n) return false;
   if (i == firstB) return false;
   return true;
}

bool checkB(string s) {
   int n = s.length();
   if (n % 2 == 0) return false;
   int i = 0, j = n - 1;
   while (i < j) {
      if (s[i] != s[j]) return false;
```

```
        i++; j--;
    }
    return true;
}

bool checkC(string s) {
    int n = s.length();
    int i = 0;
    while (i < n && s[i] == 'a') i++;
    int countA = i;
    if (countA == 0) return false;
    int j = i;
    while (j < n && s[j] == 'b') j++;
    int countB = j - i;
    if (j != n) return false;
    return countB == 2 * countA;
}

bool checkD(string s) {
    int n = s.length();
    int i = 0;
    while (i < n && s[i] == 'a') i++;
    int countA = i;
    if (countA == 0) return false;
    int j = i;
    while (j < n && s[j] == 'b') j++;
    int countB = j - i;
    if (j != n) return false;
```

```
    return countA == countB;
}
int main() {
    string s;
    cout << "Enter string: ";
    cin >> s;
    bool matched = false;
    if (checkA(s)) {
        cout << s << " belongs to Grammar 1 (S -> aS | Sb | ab)\n";
        matched = true;
    }
    if (checkB(s)) {
        cout << s << " belongs to Grammar 2 (S -> aSa | bSb | a | b)\n";
        matched = true;
    }
    if (checkC(s)) {
        cout << s << " belongs to Grammar 3 (S -> aSbb | abb)\n";
        matched = true;
    }
    if (checkD(s)) {
        cout << s << " belongs to Grammar 4 (S -> aSb | ab)\n";
        matched = true;
    }
    if (!matched)
        cout << s << " does not belong to any of the given grammars\n";
    return 0;
}
```

Output:


Problems
Output
Debug Console
Terminal
Ports
Code
PS C:\Users\Deepayan\Documents\MAVERICK\labs\5th-sem\cd-lab> cd "c:\Users\Deepayan\Documents\MAVERICK\labs\5th-sem\cd-lab\" ; if ($?) { g++ exp2.cpp -o exp2 } ; if ($?) { .\exp2 }
Enter string: aaabbb
aaabbb belongs to Grammar 1 (S -> aS | Sb | ab)
aaabbb belongs to Grammar 4 (S -> aSb | ab)
PS C:\Users\Deepayan\Documents\MAVERICK\labs\5th-sem\cd-lab>
cd "c:\Users\Deepayan\Documents\MAVERICK\labs\5th-sem\cd-lab\" ; if ($?) { g++ exp2.cpp -o exp2 } ; if ($?) { .\exp2 }
Enter string: abbbbba
abbbbba belongs to Grammar 2 (S -> aSa | bSb | a | b)
PS C:\Users\Deepayan\Documents\MAVERICK\labs\5th-sem\cd-lab>